

4. The Theory of Special Relativity states that the measurement of time for an observer moving at high speed is changed relative to a stationary observer.

(a) State the name of this effect.

1

- (b) A spacecraft is travelling at a constant speed of $2.29 \times 10^8 \text{ m s}^{-1}$ relative to a nearby planet.

A technician on the spacecraft measures a time interval to be 245 s.

Calculate the time interval as measured by an observer on the nearby planet.

3

Space for working and answer



4. (continued)

- (c) A student makes the following statement regarding special relativity:

‘Train drivers who travel at speeds of 80 m s^{-1} over large distances will have to reset their watches frequently, as they spend so many hours travelling at high speeds.’

State whether the student is correct.

Justify your answer.

2

[Turn over

